



RIVER VALLEY HIGH SCHOOL

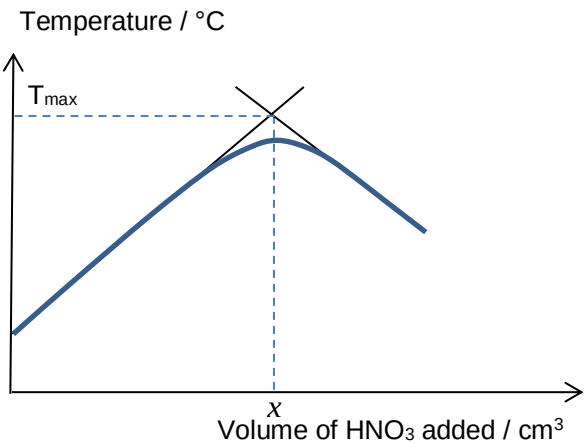
H2 CHEMISTRY

Suggested Solutions for 2014 A Level H2 Chemistry Paper**Paper 1**

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	A
2	C	22	D
3	A	23	D
4	A	24	D
5	A	25	D
6	B	26	C
7	A	27	D
8	C	28	C
9	C	29	B
10	B	30	B
11	C	31	B
12	C	32	B
13	B	33	B
14	B	34	B
15	B	35	D
16	D	36	A
17	C	37	A
18	A	38	B
19	C	39	C
20	B	40	A

Paper 2

1	(a)	$\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$	[1]
	(b)	<u>Pre-calculation</u> $\text{Ba}(\text{OH})_2 + 2\text{HNO}_3 \rightarrow \text{Ba}(\text{NO}_3)_2 + 2\text{H}_2\text{O}$ Amount of $\text{Ba}(\text{OH})_2$ used = $0.025 \times 1.00 = 0.025 \text{ mol}$ Amount of HNO_3 required to complete neutralise $\text{Ba}(\text{OH})_2 = 0.025 \times 2$ $= 0.050 \text{ mol}$ Based on the approximate concentration of HNO_3, Approximate volume of HNO_3 required at equivalence point = $0.050 / 1.5$ $= 33.33 \text{ cm}^3$ Use 50.00 cm^3 of HNO_3 in total will be used in order to determine the maximum temperature rise in the experiment. (If nitric acid is added in 5.00 cm^3 portions, this volume will give 4 data points after equivalence points.)	
		<u>Procedure and Obtaining of Results</u> 1. Fill a 50.00 cm^3 burette to 0.00 cm^3 mark with HNO_3. 2. Place a Styrofoam cup in a 250 cm^3 beaker to prevent it from toppling. Pipette 25.0 cm^3 of 1.00 mol dm^{-3} aqueous $\text{Ba}(\text{OH})_2$ into the Styrofoam cup. 3. Using a thermometer, measure and record the steady temperature of the solution, T_{initial}. Hold the thermometer when in use and do not leave it in the cup when not in use. 4. Place a lid on the Styrofoam cup to minimise heat loss to the surroundings. Make two holes in the cup to allow transfer of HNO_3 from the burette and for inserting of the thermometer. 5. Run 5.00 cm^3 of HNO_3 from the burette into the cup, stir the solution carefully with the thermometer and record the maximum temperature, T. 6. Immediately run a further 5.00 cm^3 of HNO_3 from the burette into the cup, stir and record the maximum temperature as before. 7. Continue the addition of HNO_3, in 5.00 cm^3 portions, until 50.00 cm^3 of HNO_3 have been run from the burette. 8. Record all maximum temperatures and the corresponding volume of HNO_3 added in a table. 9. Using the data from the table, plot a graph of the temperatures of the content in the Styrofoam cup against the volume of HNO_3 added.	

	<u>Recognition of Equivalence Point</u> The equivalence point will have been passed once the temperature stopped rising.	
	<u>Sketching of graph</u>  The graph comprises 2 straight lines extrapolated to intersect at the equivalence point. The first straight line is a best-fit line drawn using data before the equivalence point. As the reaction between $\text{Ba}(\text{OH})_2$ and HNO_3 is a neutralisation reaction, it is exothermic and will cause temperature to rise from the initial temperature of the $\text{Ba}(\text{OH})_2$ in the Styrofoam cup until the equivalence point where maximum temperature is achieved. The second straight line is a best-fit line drawn using data after the equivalence point. After the equivalence point, all the $\text{Ba}(\text{OH})_2$ has been neutralised so additional HNO_3 does not produce more heat. However, additional HNO_3 will lower the temperature of the reaction mixture since it has a lower temperature (rt) compared to the reaction mixture.	
	<u>Calculating the Concentration of HNO_3 and ΔH_n</u> Let $x \text{ cm}^3$ be the volume of HNO_3 added at equivalence point. Amount of $\text{Ba}(\text{OH})_2 = 25/1000 \times 1.00 = 0.025 \text{ mol}$ Amount of $\text{HNO}_3 = \text{Amount of } \text{H}_2\text{O} = 0.050 \text{ mol}$ $[\text{HNO}_3] = \frac{0.050}{\frac{x}{1000}}$ $= \frac{50}{x} \text{ mol dm}^{-3}$ Letting the maximum temperature reached during the experiment be T_{max}, maximum temperature rise is $\Delta T_{\text{max}} = T_{\text{max}} - T_{\text{initial}}$ $Q = mc\Delta T_{\text{max}} = (25 + x)(4.18)\Delta T_{\text{max}} = (104.5 + 4.18x)\Delta T_{\text{max}}$	[10]

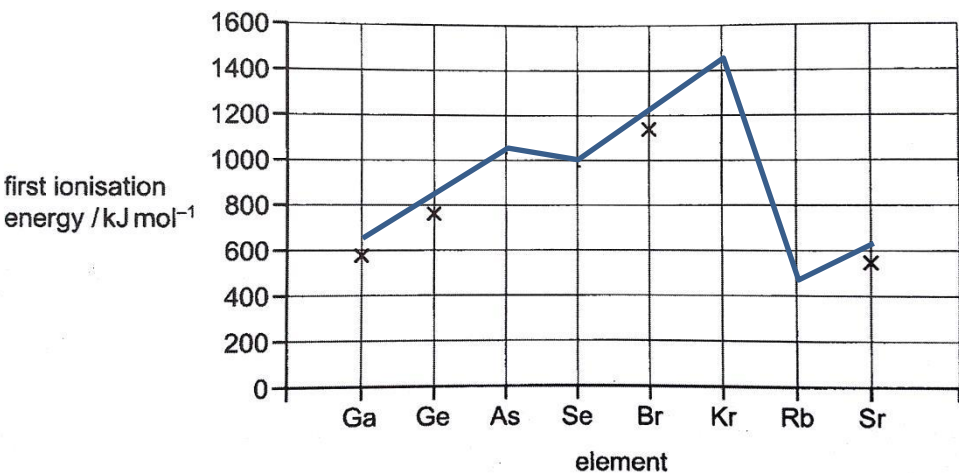
		$\Delta H_n = - \frac{Q}{(n_{H_2O \text{ produced}})}$ $= - \frac{(104.5 + 4.18x)\Delta T_{max}}{0.05} \text{ J mol}^{-1}$	
	(c)	Maximum temperature rise will not change. $2Q = (2m)c\Delta T_{max}$ $\Delta T_{max} = Q/mc$ Both the amount of heat released and the mass of reaction mixture will be doubled.	[1]

2	(a)	(i)	Mg^{2+}, SO_4^{2-} (removal of anions is by precipitation, Na^+ cannot be precipitated as all sodium salts are soluble.)	
		(ii)	Fe^{2+}, Mn^{2+}, Cu^{2+} (Zn^{2+} is not a transition metal ion.)	
	(b)		$[H^+]_{seawater} = 10^{-7.8} = 1.58 \times 10^{-8} \text{ mol dm}^{-3}$ $[H^+]_{hydrothermal \text{ vent water}} = 10^{-4.3} = 5.01 \times 10^{-5} \text{ mol dm}^{-3}$	
	(c)		$SO_4^{2-} (aq) + 10H^+(aq) + 8e^- \rightarrow H_2S(aq) + 4H_2O(l)$	
	(d)	(i)	element oxidised: sulfur (–2 to –1) element reduced: hydrogen (+1 to 0)	
		(ii)		
		(iii)	Fe^{2+} has partially filled 3d subshell ($1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$). In FeS, the 3d orbitals of iron(II) split into two sets of orbitals of different energy levels with a small energy gap. Visible light from the electromagnetic spectrum is absorbed for the promotion of a 3d electron from lower 3d energy level to higher 3d energy level. Gold yellow colour is observed because it is the complement colour of the purple light absorbed. <i>Note: candidates should make clear that the electrons are being promoted to a higher energy level with the absorption of the photon.</i>	
	(e)	(i)	The relative atomic mass is the average mass of one mole of atoms of an element relative to $\frac{1}{12}$ the mass of one mole of ^{12}C atoms.	

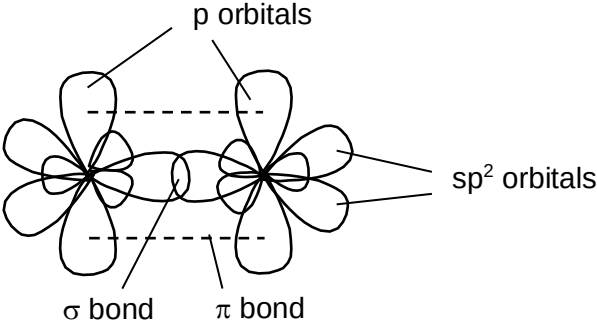
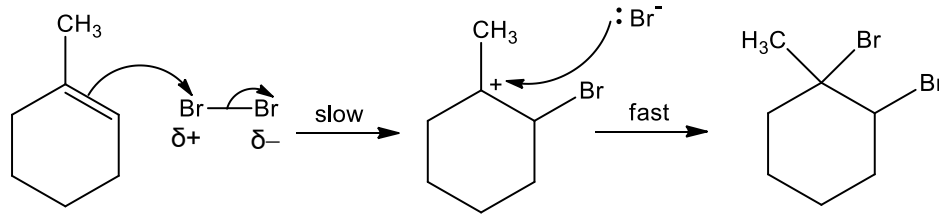
		(ii)	Let proportion of ^3He be p, ^4He is $1-p$. $4.0025959 = 3.0160293p + 4.0026033(1-p)$ $p = 7.50 \times 10^{-6}$ $\therefore 7.5 \times 10^{-4} \%$ of ^3He in the mixture.	
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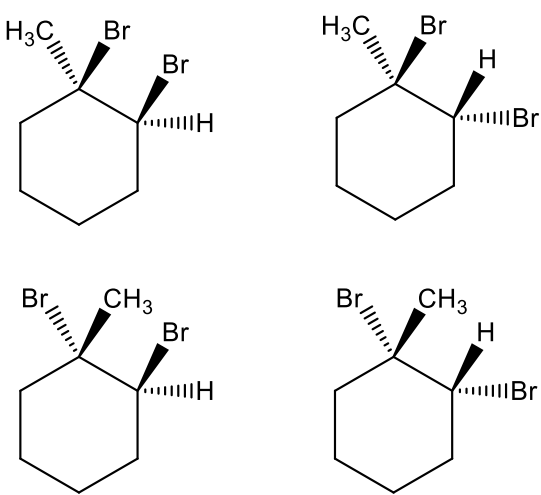
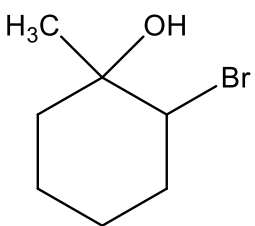
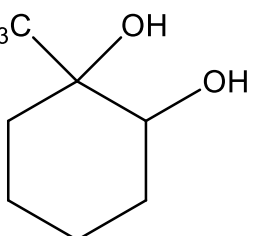
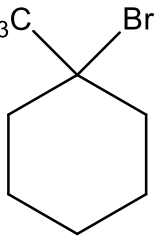
3	(a)	$pV = nRT$ $\frac{n}{V} = \frac{p}{RT}$ $\frac{n}{V} = \frac{250000}{(8.31)(25 + 273)}$ $= 100.95 \text{ mol m}^{-3}$ $= 0.101 \text{ mol dm}^{-3}$ (Note: need to account for the conversion to mol dm^{-3} as specified in the qn)	
	(b)	(i) $\frac{[\text{H}_2\text{CO}_3(\text{aq})]}{[\text{CO}_2(\text{aq})]} = K_c$ $\frac{[\text{H}_2\text{CO}_3(\text{aq})]}{[0.084]} = 1.70 \times 10^{-3}$ $[\text{H}_2\text{CO}_3(\text{aq})] = 0.084 \times 1.70 \times 10^{-3}$ $[\text{H}_2\text{CO}_3(\text{aq})] = 1.43 \times 10^{-4} \text{ mol dm}^{-3}$ (Water should not be included in the equilibrium process)	
		(ii) pH will increase. When the bottle is opened, pressure drops (from 250 kPa to 101 kPa) as CO_2 gas escapes. By Le Chatelier's Principle, the equilibrium position of reaction 1 will shift left to produce more gaseous CO_2 and thus reduces $[\text{CO}_2(\text{aq})]$. Reduction of $[\text{CO}_2(\text{aq})]$ will in turn cause the equilibrium position of reaction 2 to shift left to produce/replace more $\text{CO}_2(\text{aq})$, resulting in lower $[\text{H}_2\text{CO}_3]$. Reduction of $[\text{H}_2\text{CO}_3]$ in turn cause equilibrium position of reaction 3 to shift left, which cause $[\text{H}^+]$ to decrease and hence, pH will increase.	
	(c)	(i) Graph A 0 $t_{1/2}$	

		Note: Start concentration-time graph on the y-axis. Cannot end at zero because the reaction reached equilibrium. Graph B
	(ii)	$k = \frac{\ln 2}{t}$ $k = 23.1 \text{ s}^{-1}$
	(iii)	Let the number of half-life be n $\frac{1}{100} = \left(\frac{1}{2}\right)^n$ $n = \frac{\lg\left(\frac{1}{100}\right)}{\lg\left(\frac{1}{2}\right)}$ $n = 6.64$ Time taken = $6.64 \times 0.030 = 0.199 \text{ s}$
(d)	(i)	Graph B
	(ii)	At lower $[\text{CO}_2(\text{aq})]$ concentration, increase in $[\text{CO}_2(\text{aq})]$ will cause a proportional increase in reaction rate. As $[\text{CO}_2(\text{aq})]$ increases further, active sites of the enzymes will start becoming saturated and the increase in rate will reduce. When all the active sites are saturated, further increase of $[\text{CO}_2(\text{aq})]$ will not increase rate of reaction. Note: No need to describe how enzymes speed up reaction. This question is on variation of $[\text{CO}_2(\text{aq})]$ with rate of enzyme catalysed reaction.

4	(a)	 first ionisation energy / kJ mol^{-1} element First I. E. of As = 990 kJ mol^{-1} First I. E. of Se = 941 kJ mol^{-1} (slight dip after As) First I. E. of Kr = 1351 kJ mol^{-1} First I. E. of Rb = 403 kJ mol^{-1} (lower than Ga and Sr) Any other reasonable values that shows the correct trend.
	(b)	Ge: $[\text{Ar}]3d^{10}4s^24p^2$ As: $[\text{Ar}]3d^{10}4s^24p^3$ Se: $[\text{Ar}]3d^{10}4s^24p^4$ Br: $[\text{Ar}]3d^{10}4s^24p^5$ Kr: $[\text{Ar}]3d^{10}4s^24p^6$ Rb: $[\text{Ar}]3d^{10}4s^24p^65s^1$ From Ge to As, and Br to Kr, nuclear charge increases while shielding effect remains approximately constant. Hence, the effective nuclear charge increases and the atomic radii also decrease. Valence electrons of As and Kr are more attracted by the positive nucleus as compared to Ge and Br respectively, so more energy is required to remove them. Comparing As to Se, coulombic repulsion between the paired 4p electrons in Se makes it easier to remove one of the paired 4p electrons than an unpaired 4p electron from As. As a result, less energy is required. Less energy is required to remove the electron from the higher energy 5s subshell in Rb, as compared to removing the electron from the lower energy 4p subshell in Kr (as there is an additional inner shell of electrons in Rb and Sr). This accounts for the large drop in first I. E. between Kr and Rb.

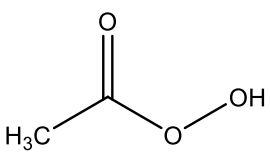
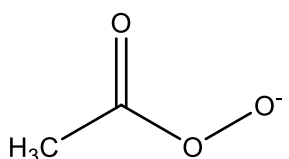
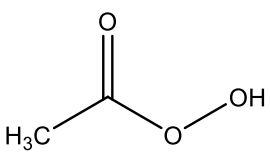
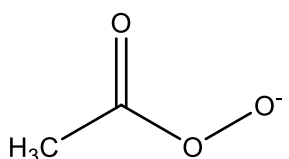
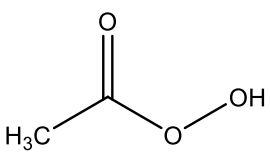
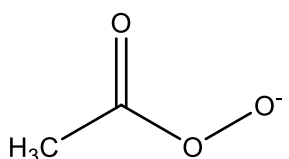
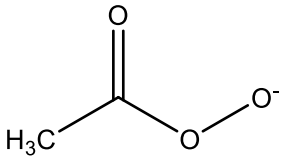
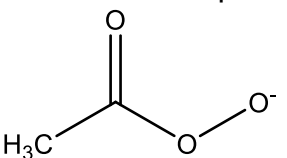
5	(a)	$\text{C}_6\text{H}_5\text{NO}_2 + 6[\text{H}] \rightarrow \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$	[2]
	(b)	$\text{C}_6\text{H}_5\text{NH}_2$ is a weak base. Under acidic conditions, $\text{C}_6\text{H}_5\text{NH}_3^+$ is formed, which is soluble in the aqueous solution as it forms ion-dipole interactions with water. The solution is made alkaline so that OH^- is available to react with $\text{C}_6\text{H}_5\text{NH}_3^+$ to form $\text{C}_6\text{H}_5\text{NH}_2$ which is immiscible with water. *Note: alkaline is not just to neutralise excess acid.	[2]
	(c)	Amount of $\text{C}_6\text{H}_5\text{NO}_2 = \frac{4.84}{123} = 0.0393 \text{ mol}$ Maximum amount of $\text{C}_6\text{H}_5\text{NH}_2 = 0.0393 \text{ mol}$ Theoretical mass of $\text{C}_6\text{H}_5\text{NH}_2 = 0.0393 \times 93.0 = 3.66 \text{ g}$ Percentage yield = $\frac{2.64}{3.66} \times 100 = 72.1 \%$	[2]
	(d)	Test: Add $\text{Br}_2(\text{aq})$ to a 1 cm^3 depth of sample in a test tube. If $\text{C}_6\text{H}_5\text{NH}_2$ is present, orange Br_2 would be decolourised, and white ppt would be formed.	[2]

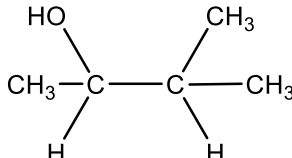
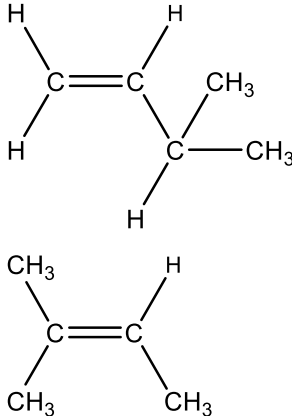
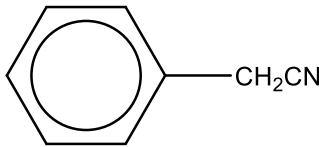
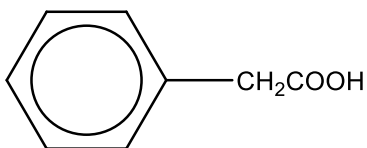
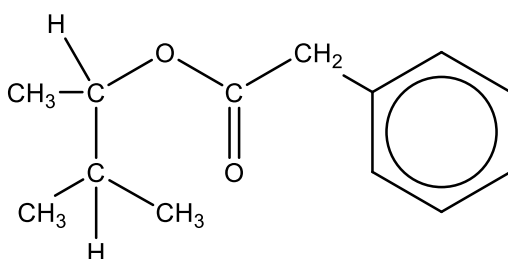
6	(a)	The C in alkenes are sp^2 hybridised. 	[2]
	(b)	(i) Electrophilic addition  * Points to note: - Positions of start and end of curly arrows - Partial charges on Br_2 and lone pair of electrons on bromide ion	[4]

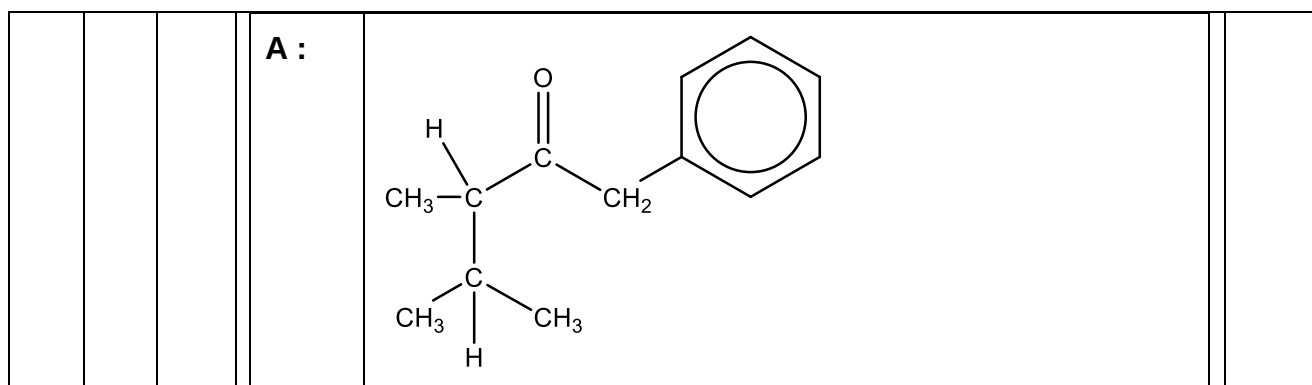
		(ii)		[2]
	(c)	 The O in H₂O has a lone pair of electrons available for donation to the carbocation. Therefore, it can behave as a nucleophile just like Br⁻. Since H₂O is present in large proportion as compared to Br⁻, the product formed is a bromo-alcohol.	[2]	
	(d)	(i)	 Step 1: cold KMnO₄(aq), NaOH(aq) Step 2: KMnO₄(aq), H₂SO₄(aq), heat under reflux	[3]
		(ii)	 Step 1: HBr(g) Step 2: excess alcoholic NH₃, heat in a sealed tube	[3]
			[Total: 16]	

Paper 3

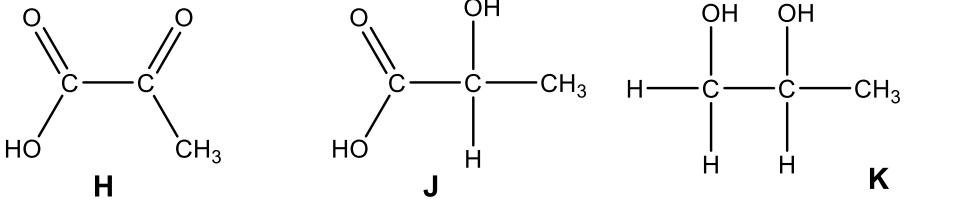
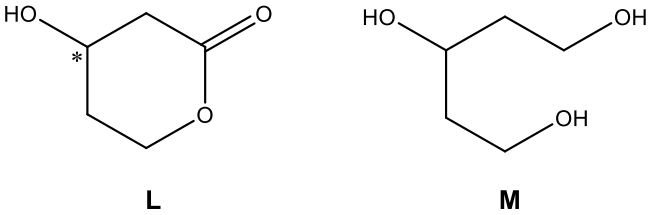
1	(a)	(i)	$\text{Na}_2\text{O(s)} + \text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)}$ pH = 13 to 14								
		(ii)	(Not in 9729 syllabus) $\text{P}_4\text{O}_6\text{(s)} + 6\text{H}_2\text{O(l)} \rightarrow 4\text{H}_3\text{PO}_3\text{(aq)}$ pH = 2 to 3								
	(b)	Amt of $\text{S}_2\text{O}_3^{2-}$ $= (16.7/ 1000) \times 0.200$ $= 3.34 \times 10^{-3} \text{ mol}$ $2 \text{ S}_2\text{O}_3^{2-} \equiv \text{Na}_2\text{O}_2$ Amt of Na_2O_2 $= 3.34 \times 10^{-3} \times 1/2$ $= 1.67 \times 10^{-3} \text{ mol}$ Mass of Na_2O $= 0.500 - (1.67 \times 10^{-3} \times M_r \text{ of } \text{Na}_2\text{O}_2)$ $= 0.370\text{g}$ Ratio $= \frac{0.370 / M_r \text{ of } \text{Na}_2\text{O}}{1.67 \times 10^{-3}}$ $= 3.57$									
	(c)	A lower $\text{p}K_a$ value means a higher K_a value and hence a stronger acid. <table border="1"><thead><tr><th>Cpd</th><th>Conjugate base</th></tr></thead><tbody><tr><td>H_2O_2</td><td>HOO^-</td></tr><tr><td>H_2O</td><td>OH^-</td></tr></tbody></table> HOO^- is more stable than OH^- as the presence of an additional electronegative O atom will disperse the negative charge on O more effectively in HOO^-. Hence, H_2O_2 is a stronger acid than H_2O.			Cpd	Conjugate base	H_2O_2	HOO^-	H_2O	OH^-	
Cpd	Conjugate base										
H_2O_2	HOO^-										
H_2O	OH^-										

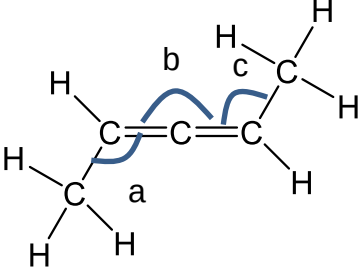
		<table><tr><th>Cpd</th><th>Conjugate base</th></tr><tr><td>CH₃COOH</td><td>CH₃COO⁻</td></tr><tr><td></td><td></td></tr></table>	Cpd	Conjugate base	CH ₃ COOH	CH ₃ COO ⁻			
Cpd	Conjugate base								
CH ₃ COOH	CH ₃ COO ⁻								
									
		CH₃COO⁻ is more stable than . The negative charge on O of CH₃COO⁻ ion can be delocalised over the -COO group as the p orbital of O atom can overlap with the π electron cloud of C=O. This is not possible in . Hence, CH₃COOH is a stronger acid.							
	(d)	(i)	Theoretical value = 350 + (2 × 360) = 1070 kJ mol⁻¹						
		(ii)	$\Delta H_r^\ominus = \Sigma BE(\text{bonds broken}) - \Sigma BE(\text{bonds formed})$ -354 = 610 - [BE(C-C) + 2 BE(C-O)] [BE(C-C) + 2 BE(C-O)] = +610 + 354 = 964 kJ mol⁻¹ Explanation: In the 3 membered ring of epoxyethane, the angle of O-C-C is <u>60°</u>. The C atom in epoxyethane is <u>sp³ hybridised</u>. Hence the optimum bond angle around each sp³ hybridised C atom is 109.5°. This will result in <u>angle strain</u> in the 3 membered ring as the electron pairs around C atom are not spaced as far apart to minimise repulsion. Therefore, the actual value is lower as the bonds are weaker.						

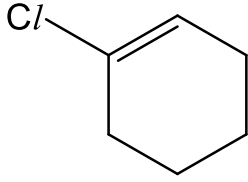
(e)	(i)	C:		
		E/ F :		
	(ii)	Limited Cl ₂ , UV light		
	(iii)			
	(iv)			
	(v)	Reaction II: ethanolic KCN, heat under reflux Reaction III: dilute H ₂ SO ₄ (aq), heat under reflux		
	(vi)	B :		



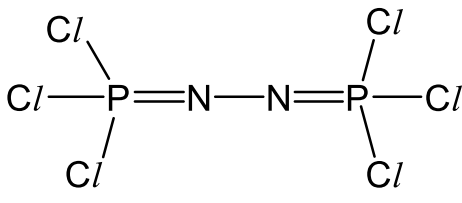
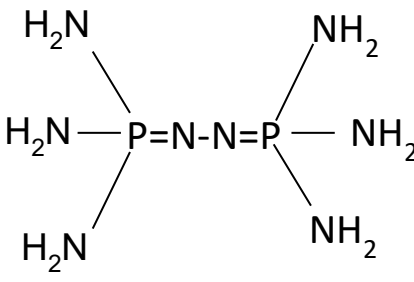
2	(a)	$ \begin{array}{c} \text{Li}^+(\text{g}) + \text{H}^-(\text{g}) \xrightarrow{\text{LE}(\text{LiH}(\text{s}))} \text{LiH}(\text{s}) \\ \begin{array}{ccc} \uparrow & & \uparrow \\ +519 & & -73. \\ \text{Li}(\text{g}) + \text{H}(\text{g}) \\ \uparrow & & \uparrow \\ +159.5 & & \frac{1}{2}(+436) \\ \text{Li}(\text{s}) + \frac{1}{2}\text{H}_2(\text{g}) \end{array} \end{array} $ $\text{Li}(\text{s}) + \frac{1}{2}\text{H}_2(\text{g}) \xrightarrow{-90.5} \text{LiH}(\text{s})$ $ \text{LE}(\text{LiH}(\text{s})) = -(+519) - (+159.5) - (-73) - \frac{1}{2}(+436) + (-90.5) \\ = -914 \text{ kJ mol}^{-1} $	
	(b)	$ \begin{array}{c} \text{AlCl}_3(\text{s}) + 4\text{LiH}(\text{s}) \xrightarrow{-276} \text{LiAlH}_4(\text{s}) + 3\text{LiCl}(\text{s}) \\ \begin{array}{ccc} \swarrow & & \searrow \\ -704 & & 4(-90.5) \\ \text{Al}(\text{s}) + \frac{3}{2}\text{Cl}_2(\text{g}) + 4\text{Li}(\text{s}) + 2\text{H}_2(\text{g}) \end{array} \end{array} $ $\text{Al}(\text{s}) + \frac{3}{2}\text{Cl}_2(\text{g}) + 4\text{Li}(\text{s}) + 2\text{H}_2(\text{g}) \xrightarrow{\Delta H_f} \text{LiAlH}_4(\text{s}) + 3\text{LiCl}(\text{s})$ $ \Delta H_f(\text{LiAlH}_4(\text{s})) + 3(-408.5) = (-704) + 4(-90.5) + (-276) \\ \Delta H_f(\text{LiAlH}_4(\text{s})) = -116.5 \text{ kJ mol}^{-1} $	

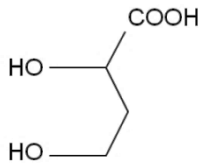
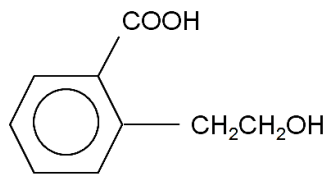
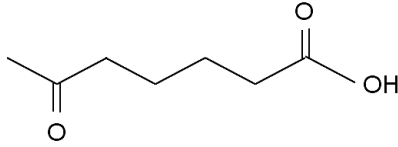
(c)	(i)	$\Delta G^{\ominus} = \Delta H^{\ominus} - T\Delta S^{\ominus}$ $\Delta S^{\ominus} = [(+3.65) - (-27.68)] \div 298 = +0.105 \text{ kJ mol}^{-1} \text{ K}^{-1}$ LiAlH₄ decomposes to give three different products, one of which is hydrogen gas. Since there is an increase in the number of gaseous molecules (and hence the number of ways the products can be arranged in the reaction vessel), $\Delta S^{\ominus}$ is positive to reflect this increased in disorder.	
	(ii)	Oxidation number of H in LiAlH₄ = -1 Oxidation number of Al in Li₃AlH₆ = +3	
	(iii)	Octahedral	
(d)	(i)	 H J K	
	(ii)	 L M	
	(iii)	L is chiral, there are four different groups bonded to chiral carbon (marked by *). The non-symmetrical ring is effectively two different groups. M is non-chiral as all carbon atoms either have two -H or -CH₂CH₂OH bonded to them.	
(e)	(i)	$\text{LiAlH}_4 + 4 \text{H}_2\text{O} \rightarrow \text{Al(OH)}_3 + 4\text{H}_2 + \text{LiOH}$ $\text{Al(OH)}_3 + 3\text{H}^+ \rightarrow \text{Al}^{3+} + 3\text{H}_2\text{O}$ $\text{LiOH} + \text{H}^+ \rightarrow \text{Li}^+ + \text{H}_2\text{O}$	
	(ii)	$n_{\text{LiAlH}_4} = \frac{2.00}{37.9} = 0.0528 \text{ mol}$ $n_{\text{H}_2} = 0.0528 \times 4 \text{ mol}$ $V_{\text{H}_2} = 0.0528 \times 4 \times 24 = 5.07 \text{ dm}^3$	

3	(a)	(Not in 9729 syllabus) Concentrated H_2SO_4 and concentrated phosphoric acid can react with NaI and NaCl to give HI and HCl respectively. $\text{NaX(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{HX(g)} + \text{NaHSO}_4(\text{aq})$ where X is I or Cl $\text{NaX(s)} + \text{H}_3\text{PO}_4(\text{aq}) \rightarrow \text{HX(g)} + \text{NaH}_2\text{PO}_4(\text{aq})$ where X is I or Cl HI, a stronger reducing agent than HCl, can reduce concentrated H_2SO_4, and itself gets oxidised to I_2. HCl cannot reduce concentrated H_2SO_4, so a pure sample of HCl can be made but not HI. $8\text{HI(g)} + \text{H}_2\text{SO}_4(\text{l}) \rightarrow \text{H}_2\text{S(g)} + 4\text{I}_2(\text{g}) + 4\text{H}_2\text{O(l)}$ However, HI is not strong enough to reduce concentrated phosphoric acid, a weaker oxidising agent than concentrated sulfuric acid. So a pure sample of HI can be made if concentrated phosphoric acid is used.
	(b) (i)	(Not in 9729 syllabus) A sample of PCl_3 can be made by heating phosphorous (P_4) with limited $\text{Cl}_2(\text{g})$. PCl_3 can be separated from reaction mixture by distillation. (Note: Heating is required for the reaction to take place and PCl_3 will be produced as a gas. Excess chlorine produces PCl_5. A distillation set up is required to recover the PCl_3 as a pure liquid)
	(ii)	(Not in 9729 syllabus) PCl_3 reacts with water vapour in the air to produce white fumes of HCl $\text{PCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_3 + 3\text{HCl}$
	(c) (i)	POCl_3
	(ii)	$\text{CH}_3\text{CH}=\text{C}=\text{CHCH}_3$ ($\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_3$ is also possible.)
	(iii)	Show displayed formula  Angle $a = c = 120^\circ$ angle $b = 180^\circ$ (For $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CCH}_3$: $\text{C}-\text{C}\equiv\text{C}$ bond angle is 180°; $\text{C}-\text{C}-\text{C}$ bond angle is 109.5°)

		(iv)	 For cyclohexane, the loss of the 2nd chlorine would result in high angle strain of the C=C=C bonds since it is not possible for the C=C=C shape to be linear in a cyclic 6-membered ring. Hence, the 2nd loss of chlorine does not occur.			
	(d)	(i)		P	N	Cl
			Mass ratio	20.5	9.2	70.3
			Mole ratio	$\frac{20.5}{31} = 0.661$	$\frac{9.2}{14} = 0.657$	$\frac{70.3}{35.5} = 1.98$
			Simplest ratio	1	1	3

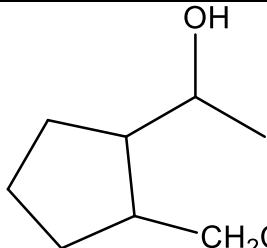
Empirical formula: **PNCI₃**

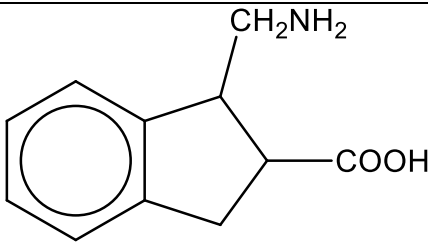
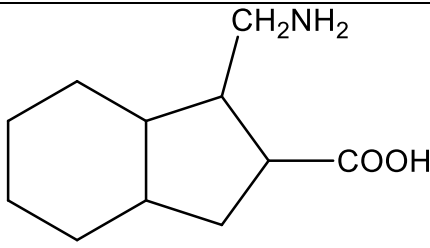
			Molecular formula: P ₂ N ₂ Cl ₆ Structure: 			
		(ii)	$12\text{NH}_3 + \text{P}_2\text{N}_2\text{Cl}_6 \rightarrow \text{P}_2\text{H}_{12}\text{N}_8 + 6\text{NH}_4\text{Cl}$			
		(iii)	Compound T: 			

4	(a)	(i)	The electron-withdrawing Cl atom disperses the negative charge on COO^- in the $\text{ClCH}_2\text{COO}^-$ anion, giving rise to a more stable anion compared to CH_3COO^-. When anion is more stable, the equilibrium position of acid dissociation $\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$ lies more to the right giving rise to larger extent of dissociation and greater K_a. Hence, ClCH_2COOH is a stronger acid with a lower $\text{p}K_a$ compared to CH_3COOH.
		(ii)	For CH_3COOH, $\text{pH} = \text{p}K_a + \lg [\text{anion}]/[\text{acid}]$ $3.8 = 4.76 + \lg [\text{anion}]/[\text{acid}]$ $[\text{anion}]/[\text{acid}] = 0.110$ For ClCH_2COOH, $\text{pH} = \text{p}K_a + \lg [\text{anion}]/[\text{acid}]$ $3.8 = 2.87 + \lg [\text{anion}]/[\text{acid}]$ $[\text{anion}]/[\text{acid}] = 8.51$
	(b)	(i)	Step 1: dilute $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux Step 2: $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux
			U: 
		(ii)	Step 1: $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux Step 2: $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux
			V: 
		(iii)	Step 1: $\text{KMnO}_4(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux Step 2: I_2, $\text{NaOH}(\text{aq})$, warm, followed by $\text{H}_2\text{SO}_4(\text{aq})$
			W: 

	(c)	(i)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{COO}^- \\ \\ \text{H} - \text{C} - \text{CH}_3 \\ \\ \text{CH}_3 \end{array} $
		(ii)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{C}_6\text{H}_4 \\ \\ \text{OH} \end{array} $
		(iii)	$ \begin{array}{c} \text{H} \\ \\ \text{H}_3\text{N}^+ - \text{C} - \text{COO}^- \\ \\ \text{CH}_2 \\ \\ \text{COO}^- \end{array} $
	(d)	(i)	(Not in 9729 syllabus) It is likely to interact with the acidic side chain of aspartic acid through ionic interactions.
		(ii)	(Not in 9729 syllabus) It is likely to interact with the non-polar side chain of valine through van der Waals' forces.

5	(a)	(i)	$\text{Cu}^{2+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$	[1]	
		(ii)	$\text{Ni}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$	[1]	
	(b)	A blue ppt of $\text{Cu}(\text{OH})_2$ is formed when aqueous ammonia is first added to a solution of Cu^{2+}. Ammonia reacts with water to produce OH^- ions which in turn precipitate the Cu^{2+}. $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$ $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s}) + 6\text{H}_2\text{O}(\text{l})$ Blue ppt dissolves in excess of $\text{NH}_3(\text{aq})$ to produce $[\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}$, giving a dark blue solution. $\text{Cu}(\text{OH})_2(\text{s}) + 4\text{NH}_3(\text{aq}) + 2\text{H}_2\text{O}(\text{l}) \rightleftharpoons [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$			[4]
	(c)	(i)	$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni} \quad E^\ominus = -0.25 \text{ V}$ $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu} \quad E^\ominus = +0.34 \text{ V}$ $E^\ominus_{\text{cell}} = E^\ominus(\text{Cu}^{2+}/\text{Cu}) - E^\ominus(\text{Ni}^{2+}/\text{Ni})$ $= +0.34 - (-0.25)$ $= +\underline{0.59 \text{ V}} > 0$ (energetically feasible)	[1]	
		(ii)	When $[\text{Ni}^{2+}]$ is reduced, the equilibrium position of $\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$ shifts to the left to produce more Ni^{2+}, resulting in $E(\text{Ni}^{2+}/\text{Ni})$ becoming <u>more negative</u>. Hence E_{cell} becomes <u>more positive</u> than +0.59 V. <u>*Note: do not use ambiguous terms like 'higher' or 'lower'.</u>	[1]	
	(d)	$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni} \quad E^\ominus = -0.25 \text{ V}$ $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu} \quad E^\ominus = +0.34 \text{ V}$ $\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag} \quad E^\ominus = +0.80 \text{ V}$ Since $E^\ominus(\text{Ni}^{2+}/\text{Ni})$ is more negative than $E^\ominus(\text{Cu}^{2+}/\text{Cu})$, Ni (and Cu too) would be <u>oxidised</u> to Ni^{2+} at the anode. However, at the cathode, Cu^{2+} is preferentially reduced. So Ni^{2+} would remain dissolved in the solution. Since $E^\ominus(\text{Ag}^+/\text{Ag})$ is more positive than $E^\ominus(\text{Cu}^{2+}/\text{Cu})$, hence Ag will not be <u>oxidised</u> at the anode but it will fall off the anode as anode sludge during electrolysis. <u>*Note that question requires candidates to explain specifically what happens to both Ni and Ag during the entire purification process. Hence it is not sufficient to just explain which metal is preferentially oxidised at the anode.</u>			[3]

(e)	(i)	$Q = It = 2.00 \times 23.0 \times 60 = 2760 \text{ C}$ $\text{Amount of electrons} = \frac{2760}{96500} = 2.86 \times 10^{-2} \text{ mol}$ $\text{Amount of Cu reduced from Cu}^{2+} = \frac{2.86 \times 10^{-2}}{2} = 1.43 \times 10^{-2} \text{ mol}$ $\text{Expected increase in mass of cathode} = 1.43 \times 10^{-2} \times 63.5$ $= \underline{\underline{0.908 \text{ g}}}$	[2]														
	(ii)	$\text{Mass of Ag lost} = \underline{\underline{0.0500 \text{ g}}}$ $\text{Amount of Ni(C}_4\text{H}_7\text{N}_2\text{O}_2)_2 = \frac{0.492}{58.7+2(115)} = 1.70 \times 10^{-3} \text{ mol}$ $= \text{amount of Ni lost from anode}$ $\text{Mass of Ni lost} = 1.70 \times 10^{-3} \times 58.7 = \underline{\underline{0.100 \text{ g}}}$ $\text{Mass of Cu lost} = 0.950 - 0.0500 - 0.100 = \underline{\underline{0.800 \text{ g}}}$	[3]														
(f)	<table><tr><th>Reaction</th><th>Deduction</th></tr><tr><td>X and Y can undergo <u>acid-metal reaction</u> with Na, only Y can undergo <u>acid-carbonate reaction</u> with Na_2CO_3.</td><td>X is an <u>alcohol</u>. Y is a <u>carboxylic acid</u>.</td></tr><tr><td>X and Y cannot undergo <u>condensation</u> with 2,4-DNPH.</td><td>X and Y <u>do not contain aldehydes or ketones</u>.</td></tr><tr><td>X and Y cannot undergo <u>electrophilic addition</u> with $\text{Br}_2(\text{aq})$.</td><td>X and Y <u>do not contain alkene and alkyne functional groups</u>.</td></tr><tr><td>Z and Ω can undergo <u>acid-base reaction</u> with HCl to form salt.</td><td>Z and Ω are <u>amines</u>.</td></tr><tr><td>Z and Ω can undergo <u>acid-carbonate reaction</u> with Na_2CO_3 to form salt.</td><td>Z and Ω are <u>carboxylic acids</u>.</td></tr><tr><td>Z has <u>6 H less</u> than Ω.</td><td>Z contains a <u>benzene</u>, whereas Ω contains a <u>cyclohexane</u>.</td></tr></table>  X: Y: $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$		Reaction	Deduction	X and Y can undergo <u>acid-metal reaction</u> with Na, only Y can undergo <u>acid-carbonate reaction</u> with Na_2CO_3 .	X is an <u>alcohol</u> . Y is a <u>carboxylic acid</u> .	X and Y cannot undergo <u>condensation</u> with 2,4-DNPH.	X and Y <u>do not contain aldehydes or ketones</u> .	X and Y cannot undergo <u>electrophilic addition</u> with $\text{Br}_2(\text{aq})$.	X and Y <u>do not contain alkene and alkyne functional groups</u> .	Z and Ω can undergo <u>acid-base reaction</u> with HCl to form salt.	Z and Ω are <u>amines</u> .	Z and Ω can undergo <u>acid-carbonate reaction</u> with Na_2CO_3 to form salt.	Z and Ω are <u>carboxylic acids</u> .	Z has <u>6 H less</u> than Ω .	Z contains a <u>benzene</u> , whereas Ω contains a <u>cyclohexane</u> .	[4]
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			[Total: 20]	